

Cloud Native Network Observability

Treating Routers like Kubernetes Resources

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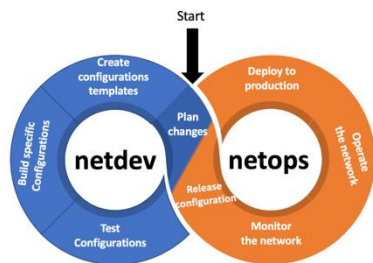
About me

NetDevOps
Engineer

Co-Founder of
NetAuto
Group

Masterstudies
in Computer
Science at H-
DA

Crossfit and
Running





Intelligent
Connectivity



Hybrid Cloud



Data & AI



Security



Digital Workspace



Smart Infrastructure
Engineering

Selection of our Key Alliances



Certifications



Total Employees

600+

Revenue

350M+

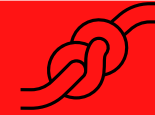


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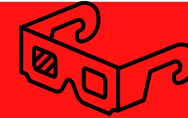
- ▶ **Background**
- ▶ **Solution**
- ▶ **Demo**
- ▶ **Outro**

Where are we coming from?

The "Pull" Bottleneck



Dangerous Blind Spots



Fragile Data Parsing



Static and Siloed Infrastructure





Observability, that shows us in real time



Standardized Data Model



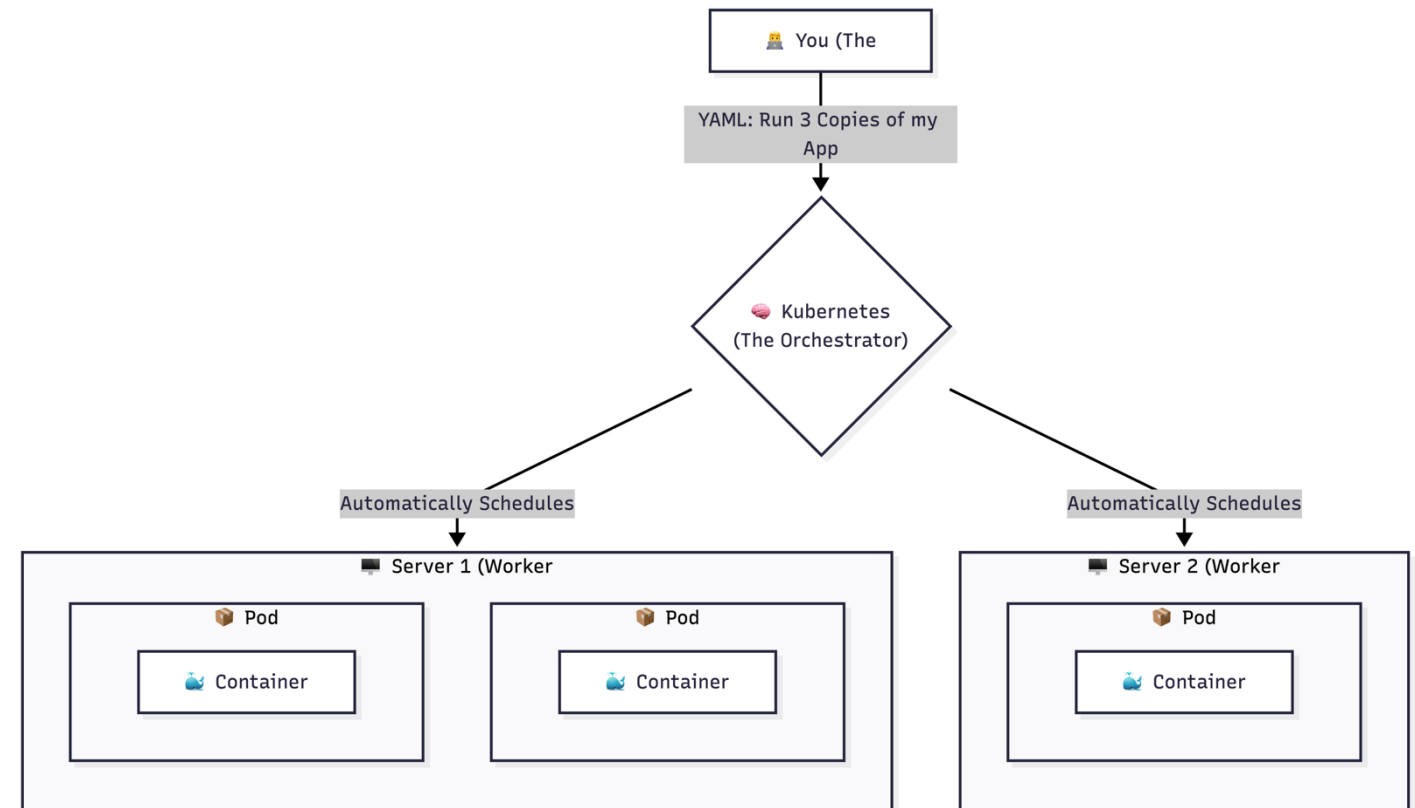
Scalability



Bridging the gap

What is Kubernetes?

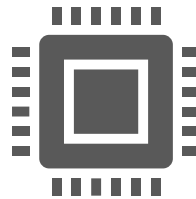
- ▶ **Declarative Configuration**
- ▶ **Self Healing**
- ▶ **Scalability**
- ▶ **Extendability**





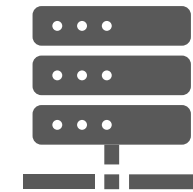
Created in 1980

Grandfather of network monitoring



Pull model

Monitoring server asks the device every time
For example: What is the byte count on interface 1?



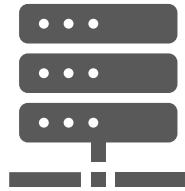
Data

Stored as MIBs(Management Information)
OID (Object Identifiers) like
1.3.6.1.2.1.2.2.1.10.1

NETCONF



Created in mid-2000s



**SSH-Tunnel to securely send
XML-encoded messages**



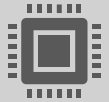
YANG Model

interfaces/interface/state/counters/in-
octets



Developed by the
OpenConfig working group

heavily driven by Google



Uses gRPC

Subscribes to data path in case, instead of
asking for it all the time



YANG model

Like NETCONF, but uses
Protocolbuffers or JSON to transport
data



Why gNMI Wins the Telemetry War

Speed

• *Pull* → **Push (Streaming)**

Encoding

• *Text/XML* → **Binary (Protobuf)**

Transport

• *Single-thread* → **HTTP/2 Multiplexed**

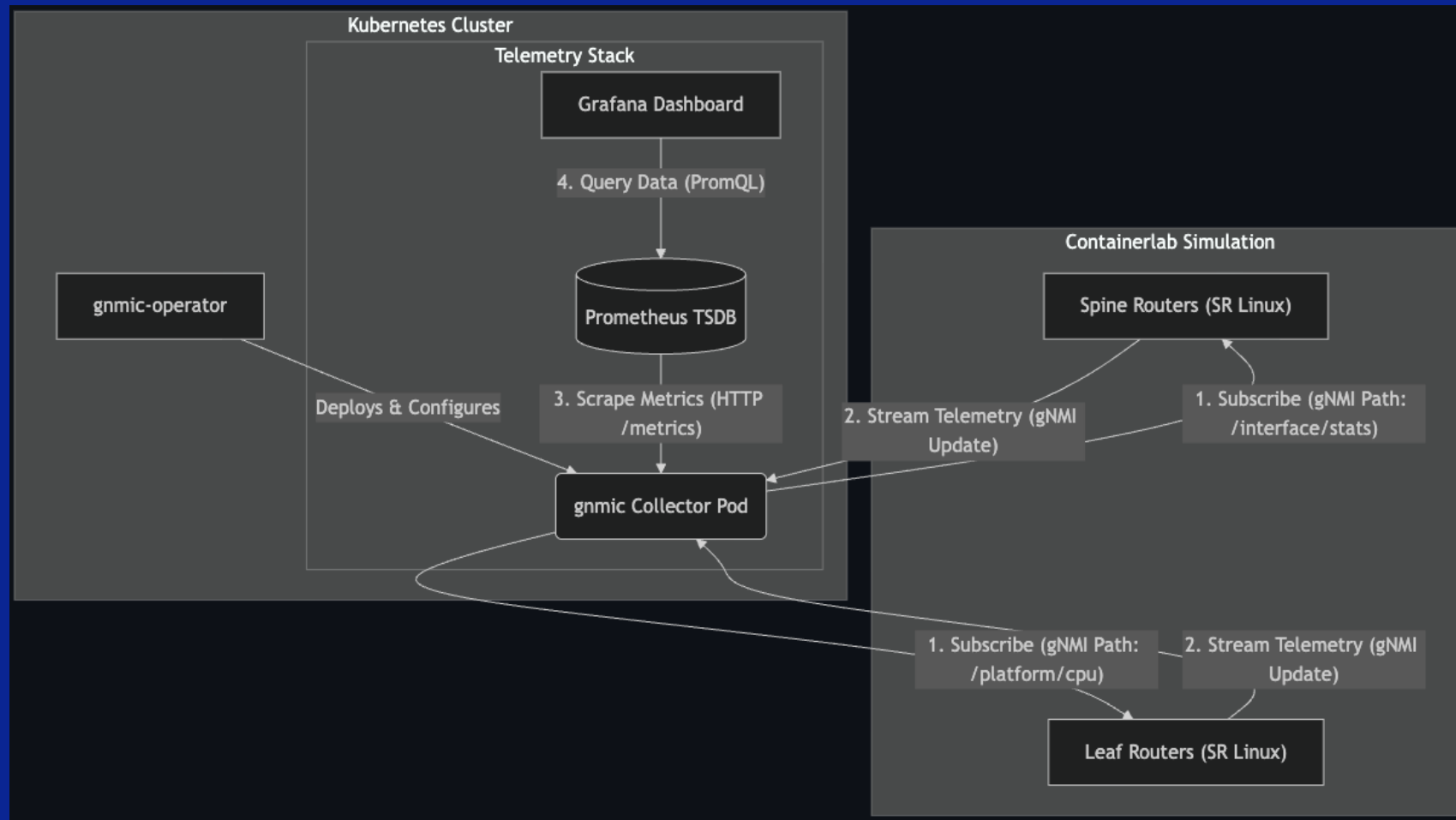
Ecosystem

• *Siloed VMs* → **Kubernetes Native**

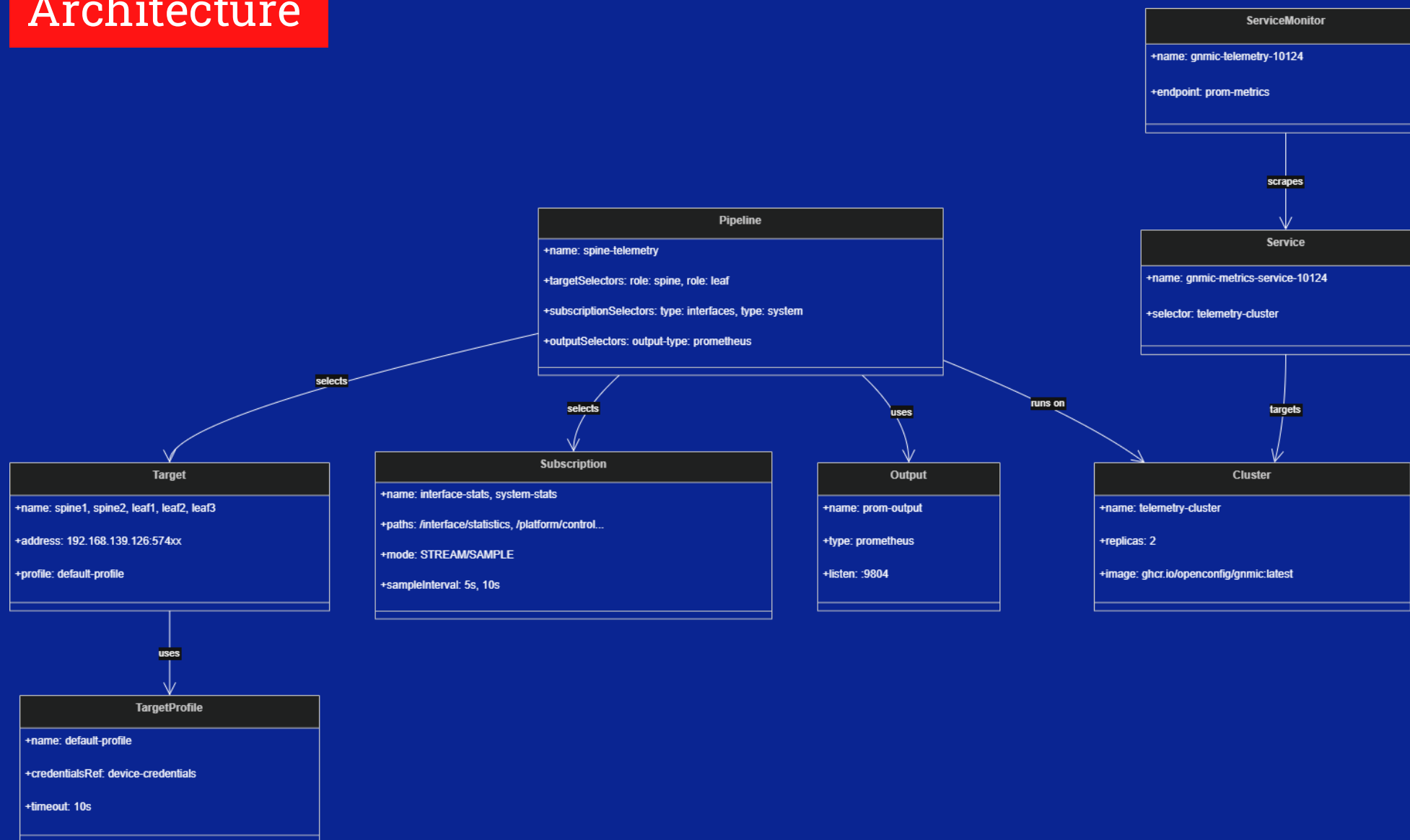


Solution

Setup



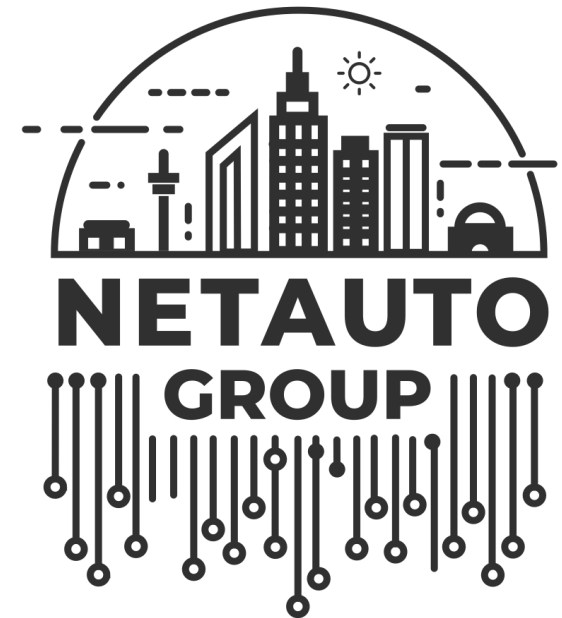
Architecture





Demo

- Network Automation enthusiasts
- Meetup every two months
- Join us for more interesting topics!
- Next Meetup: 09.04.2026
- Location: Logicalis Office, Siemensstraße 10, Neu-Isenburg



Questions?

- ▶ Coming from SNMP and single VM collection platform
- ▶ Kubernetes as platform for Observability
 - I. Selfhealing
 - II. Scalable
 - III. Declarative
- ▶ GNMIC-Operator for connecting to the devices
- ▶ Collecting the data in prometheus
- ▶ Showing it in Grafana



GitHub Repository
for the Talk

Thank You and Have a Great Day!

Let's stay connected!



de.logicalis.com



Join us on LinkedIn

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